

CR-001: Extended System Information

What Changed and Why

Addition	Reason
Airline	Flights had no record of which airline operates them
Gate	No tracking of which gate a flight departs from
Baggage	Booking baggage was not tracked individually
FlightDelayLog	Delay reasons and durations were not recorded

Existing Tables Modified

Flight table - 2 new columns:

- `airline_id` : INT NOT NULL — FK to Airline
- `gate_id` : INT NULL — FK to Gate

Challenges When Altering Existing Tables

Adding NOT NULL column to Flight which already has 8 rows. SQL Server does not allow adding NOT NULL with no default to existing data.

Solution:

1. Add column as NULL
2. Update all rows with a valid value
3. Change column to NOT NULL

Referential Actions Justification

FK	ON DELETE	Reason
Flight.airline_id	NO ACTION	Flights are historical. Deleting airline must not delete flights.
Flight.gate_id	SET NULL	If gate deleted, flight stays with NULL gate.
Gate.airport_id	CASCADE	Gates belong to airports. No airport = no gate.
Baggage.booking_id	CASCADE	Baggage only exists because of a booking.
FlightDelayLog.flight_id	CASCADE	Delay logs only exist because of a flight.

Most Complex Query

Final Challenge (Advanced Q6) joins 7 tables Uses LEFT JOIN for Gate Booking Baggage FlightDelayLog Uses COALESCE to return 0 instead of NULL Uses ISNULL to show No Gate when gate_id is NULL Uses COUNT DISTINCT to avoid duplicates from multiple joins Orders by TotalRevenue DESC